CHAPTER 5

BIODIVERSITY AND CONSERVATION

CHAPTER ESSENCE AT A GLANCE:
Latitudinal diversity gradients, species-area log curves, anthropogenic extinction threats, and in-situ versus ex-situ bio-reserves.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Levels of Biodiversity and Latitudinal Gradients:** Biodiversity exists at three levels: genetic diversity (Rauwolfia vomitoria reserpine potency, 50,000 rice strains in India), species diversity (Western Ghats have greater amphibian diversity than Eastern Ghats), and ecological diversity (rainforests, mangroves, coral reefs). Latitudinal gradient: species richness peaks at the equator and declines toward the poles because tropics remained undisturbed by glaciations, receive constant solar energy, and experience less seasonality.

• **Alexander von Humboldt Species-Area Model:** Alexander von Humboldt observed that species richness increases with explored area, but only up to a limit. Described by a rectangular hyperbola on arithmetic scale, and a straight line on logarithmic scale: log S = log C + Z log A (S = species richness, A = area, C = Y-intercept, Z = slope or regression coefficient). For local regions, Z ranges from 0.1 to 0.2; across entire continents, Z steepens significantly to 0.6 to 1.2 (e.g., frugivorous birds in tropical forests Z = 1.15).

• **The Evil Quartet: Anthropogenic Extinction Drivers:** Accelerated biodiversity loss is driven by four primary causes (The Evil Quartet): (1) Habitat loss and fragmentation (clearing of Amazon rainforest lungs of our planet), (2) Over-exploitation (Steller's sea cow, passenger pigeon), (3) Alien species invasions (Nile perch introduced into Lake Victoria extinguishing >200 cichlid fish species; weeds like Lantana, Eichhornia water hyacinth, Parthenium; African catfish Clarias gariepinus), and (4) Co-extinctions (extinction of host leads to extinction of obligate mutualists).

• **In-Situ versus Ex-Situ Conservation Paradigms:** In-situ (on-site conservation in natural habitat): National Parks (strict protection, no exploitation), Wildlife Sanctuaries, Biosphere Reserves (14 in India), Sacred Groves (Khasi and Jaintia hills in Meghalaya, Aravalli hills in Rajasthan - last refuges for rare plants), and Biodiversity Hotspots (high endemism and threat: Western Ghats-Sri Lanka, Indo-Burma, Himalaya). Ex-situ (off-site protection): Zoological parks, Botanical gardens, Wildlife safari parks, Cryopreservation of gametes at -196°C in liquid nitrogen, and Seed banks.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\log S = \log C + Z \log A \quad [S = \text{Richness},\ A = \text{Area},\ Z = \text{Slope},\ C = Y\text{-intercept}]$$

Alexander von Humboldt Logarithmic Species-Area Formulation

$$\text{Slope Constants} = Z = 0.1 \text{ to } 0.2 \text{ (Local regions)} \Longleftrightarrow Z = 0.6 \text{ to } 1.2 \text{ (Entire continents)}$$

Ecological Regression Slope Scaling by Spatial Area

$$\text{The Evil Quartet} = \text{Habitat Loss} + \text{Over-exploitation} + \text{Alien Invasions} + \text{Co-extinctions}$$

Four Primary Anthropogenic Drivers of Accelerated Extinction

$$\text{Conservation Modalities} = \frac{\text{In-Situ: National Parks, Biosphere Reserves, Sacred Groves}}{\text{Ex-Situ: Cryopreservation } (-196^\circ\text{C}),\ \text{Seed Banks, Botanical Gardens}}$$

On-Site Ecosystem Protection vs Off-Site Germplasm Banking

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Environmental Issues (Air and Water Pollution, Solid Waste, Global Warming, Ozone Depletion) was completely deleted under the rationalized syllabus.

• **Alert:** The regression coefficient Z of the species-area curve is 0.1 to 0.2 for normal local regions, but steepens to 0.6 to 1.2 when analyzing entire continental landmasses.

• **Alert:** Sacred groves are an IN-SITU conservation method (not ex-situ!) where forest tracts are protected by indigenous communities due to cultural and religious beliefs.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is Alexander von Humboldt's Species-Area Relationship? Write its logarithmic equation and define each variable. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Relationship: Within a region, species richness increases with increasing explored area, but only up to a limit; plotted on a logarithmic scale, it yields a straight line. **[1 Mark]**

• Equation: log S = log C + Z * log A, where S = Species richness, A = Area, Z = Slope of the line (regression coefficient), and C = Y-intercept. **[1 Mark]**

2. Distinguish between in-situ conservation and ex-situ conservation with two specific examples for each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• In-situ (on-site) conservation: Protecting endangered species in their natural habitats where ecological and evolutionary processes continue (e.g., National Parks, Biosphere Reserves, Sacred Groves). **[1 Mark]**

• Ex-situ (off-site) conservation: Taking threatened animals and plants out from their natural habitat and placing them in human-controlled protective settings (e.g., Zoological parks, Botanical gardens, Cryopreservation seed banks). **[1 Mark]**

3. Why do tropical regions exhibit greater biological diversity than temperate and polar regions? Explain three major ecological hypotheses. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Speciation time: Temperate regions experienced frequent glaciations in the past, whereas tropical latitudes remained relatively undisturbed for millions of years, allowing long evolutionary speciation. **[1 Mark]**

• Less seasonal environment: Tropical climates are less seasonal, more constant, and predictable, promoting niche specialization and greater species richness. **[1 Mark]**

• Higher solar energy: Tropics receive more direct solar radiation, contributing to higher photosynthetic primary productivity, supporting larger food webs and diverse species. **[1 Mark]**

4. Explain 'The Evil Quartet' - the four major anthropogenic causes driving accelerated species extinction across the globe. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Habitat loss and fragmentation: Most important driver (e.g., clearing of Amazon rainforest for soybean cultivation; large habitats broken into small fragments isolate species). **[0.75 Mark]**

• Over-exploitation: Excessive harvesting by humans turning need into greed has led to the extinction of species like Steller's sea cow and passenger pigeon. **[0.75 Mark]**

• Alien species invasions: Introduction of non-native species that turn invasive, outcompeting native wildlife (e.g., Nile perch introduced into Lake Victoria). **[0.75 Mark]**

• Co-extinctions: When a host species becomes extinct, any obligate mutualist or parasite tied to it also faces mandatory extinction (e.g., coevolved plant-pollinator mutualisms). **[0.75 Mark]**

5. Discuss the ecological menace caused by alien invasive species, citing the examples of Nile perch in Lake Victoria, Eichhornia crassipes, and the African catfish Clarias gariepinus. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Nile perch in Lake Victoria: Introduced into Lake Victoria in East Africa, this predatory fish led to the ecological extinction of an entire endemic assemblage of more than 200 species of cichlid fishes. **[1 Mark]**

• Eichhornia crassipes (water hyacinth): Introduced for its lovely flowers, it grew invasively, choking water bodies, depleting dissolved oxygen, and causing massive death of aquatic life. **[1 Mark]**

• Clarias gariepinus (African catfish): Illegally introduced for aquaculture purposes, it poses a severe threat to indigenous indigenous catfishes in Indian rivers. **[1 Mark]**

6. What are Sacred Groves? Name two regions in India where sacred groves are found, and explain their significance in conserving rare and threatened plants. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Definition: Tracts of pristine forest that are set aside, where all trees and wildlife are venerated and given total community protection due to religious and cultural traditions. **[1 Mark]**

• Indian locations: Khasi and Jaintia Hills in Meghalaya, Aravalli Hills in Rajasthan, Western Ghats regions of Karnataka and Maharashtra, and Sarguja, Chanda, and Bastar areas of Madhya Pradesh. **[1 Mark]**

• Conservation significance: In Meghalaya, sacred groves represent the last remaining refuges for a large number of rare, endemic, and threatened floral and faunal species that have disappeared elsewhere. **[1 Mark]**

7. A remote biodiversity hotspot faces habitat fragmentation due to highway construction, which will divide a contiguous forest of 10,000 square kilometers into twenty small fragmented patches of 500 square kilometers each. Predict the ecological impact on large apex mammals, explain the effect on the regression coefficient Z in the species-area curve, and propose an in-situ mitigation plan. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Ecological impact on apex mammals: Mammals and birds requiring large home ranges and animals with migratory habits are adversely affected; small isolated populations suffer inbreeding depression and local extinctions. **[1.5 Marks]**

• Effect on regression coefficient Z: In fragmented small areas, local species richness drops rapidly; across the landscape, the slope Z becomes significantly steeper as habitat loss isolates species and reduces diversity. **[1.5 Marks]**

• In-situ mitigation plan: Construct elevated wildlife flyovers and eco-ducts (canopy bridges and underpasses) across the highway to maintain contiguous habitat corridors, reconnecting the fragmented patches into a unified genetic reserve. **[1 Mark]**